# FIITJEE
# ALL INDIA TEST SERIES
## JEE (Advanced)-2024
## FULL TEST – IX
## PAPER –1
## TEST DATE: 12-05-2024

**Time Allotted: 3 Hours** **Maximum Marks: 180**

**General Instructions:**

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

  **Section – A *(01 –03, 18 – 20, 35 – 37):*** This section contains **NINE (9)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

  **Section – A *(04 – 07, 21 – 24, 38 – 41)*:** This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

  **Section – A *(08 – 11, 25 – 28, 42 – 45):*** This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (P), (Q), (R) and (S) and **FIVE** statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

  **Section – B *(12 – 17, 29 – 34, 46 – 51):*** This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

**MARKING SCHEME**

**Section – A (One or More than One Correct):** Answer to each question will be evaluated according to the following marking scheme:

| | | | |
|---|---|---|---|
| Full Marks | : | +4 | If only (all) the correct option(s) is (are) chosen; |
| Partial Marks | : | +3 | If all the four options are correct but ONLY three options are chosen; |
| Partial marks | : | +2 | If three or more options are correct but ONLY two options are chosen and both of which are correct; |
| Partial Marks | : | +1 | If two or more options are correct but ONLY one option is chosen and it is a correct option; |
| Zero Marks | : | 0 | If none of the options is chosen (i.e. the question is unanswered); |
| Negative Marks | : | –2 | In all other cases. |

**Section – A (Single Correct):** Answer to each question will be evaluated according to the following marking scheme:

| | | | |
|---|---|---|---|
| *Full Marks* | : | +3 | If ONLY the correct option is chosen. |
| Zero Marks | : | 0 | If none of the options is chosen (i.e. the question is unanswered); |
| *Negative Marks* | : | –1 | In all other cases. |

**Section – B:** Answer to each question will be evaluated according to the following marking scheme:

| | | | |
|---|---|---|---|
| *Full Marks* | : | +4 | If ONLY the correct numerical value is entered at the designated place; |
| Zero Marks | : | 0 | In all other cases. |

# Physics

## PART – I

### SECTION – A
### (One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

1. The pitch of screw gauge is 1 mm and its circular scale is divided into 100 divisions. When nothing is put between the studs the zero of main scale is not seen. But when circular scale is rotated by 225° the zero of main scale is seen and the zero of main scale coincide with the zero of circular scale. When a glass plate is placed between the studs, the circular scale lies slightly right to the $15^{th}$ division of main scale and circular scale reads 18 divisions. Then,
   (A) Error is positive zero error and its magnitude is 0.375 mm.
   (B) Error is negative zero error and its magnitude is 0.625 mm.
   (C) The thickness of the glass plate is 14.555 mm.
   (D) The thickness of the glass plate is 15.805 mm.

2. A cubical box of side 20 m contains helium gas at a pressure 160 $N/m^2$. During an observation time of 10 second, an atom travelling with root mean square speed parallel to one of the sides of the cube was found to make 1000 hits with a particular wall, without any collision with the other atoms. (take $R = \frac{25}{3}$ J/mol-K). Then
   (A) The temperature of gas is 2560 K.
   (B) The temperature of gas is 5120 K.
   (C) The total mass of helium gas in the box is 240 gm.
   (D) The total mass of helium gas in the box is 120 gm.

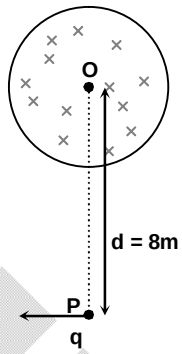
3. A neutron with kinetic energy 60 eV collides inelastically with a singly ionized helium atom at rest. After collision, the neutron is scattered at right angle to the original motion direction and singly ionized helium atom is in first excited state. Then choose the correct option(s)
   (A) The kinetic energy of neutron after collision is 3.36 eV
   (B) The kinetic energy of neutron after collision is 6.36 eV
   (C) The kinetic energy of $He^+$ atom after collision is 15.84 eV
   (D) The kinetic energy of $He^+$ atom after collision is 12.84 eV.

### SECTION – A
### (One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

4. A voltmeter of resistance $R_1$ and an ammeter of resistance $R_2$ are connected in series across a battery of emf V = 900 volts and having internal resistance 10Ω. When a resistance 10Ω is connected in parallel to voltmeter, reading of ammeter increases eight times while that of voltmeter reduces to half. Then, the value of resistor $R_2$ is
   (A) $\frac{2}{7}\Omega$ (B) $\frac{7}{5}\Omega$
   (C) $\frac{7}{2}\Omega$ (D) $\frac{5}{7}\Omega$

5. In the cylindrical region of radius R = 4m, there exists a time varying magnetic field B such that $\frac{dB}{dt} = K$ Tesla/sec, where K is a positive constant. A charged particle having charge q = 10C is placed at the point P at a distance d from its center O. Now, the particle is moved in the direction perpendicular to line OP by an external agent up to 8 m so that the work done by external agent is $80\pi$ Joule and there is no gain in the kinetic energy of charged particle. Then, the value of K is

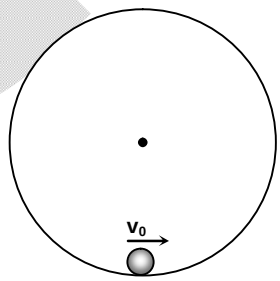



(A) 2 (B) 4
(C) 6 (D) 8

6. A satellite of mass m is orbiting about a planet in a circular orbit of radius 3R where R is the radius of the planet. It starts losing its mechanical energy due to small air resistance at the rate of 10 J/s. The time taken by the satellite to hit the surface of the planet is

(A) $\frac{GMm}{10R}$ (B) $\frac{GMm}{20R}$
(C) $\frac{GMm}{30R}$ (D) $\frac{GMm}{40R}$

7. A ring of mass 6m is kept on a smooth horizontal surface with its face horizontal. The radius of hollow cylinder is R. A small ball of mass 4m is also kept very close to the circumference of ring as shown in figure. Now, a velocity $v_0$ is given to the ball towards right. The normal force between small ball and ring at t = 0 is (neglect the gravity)

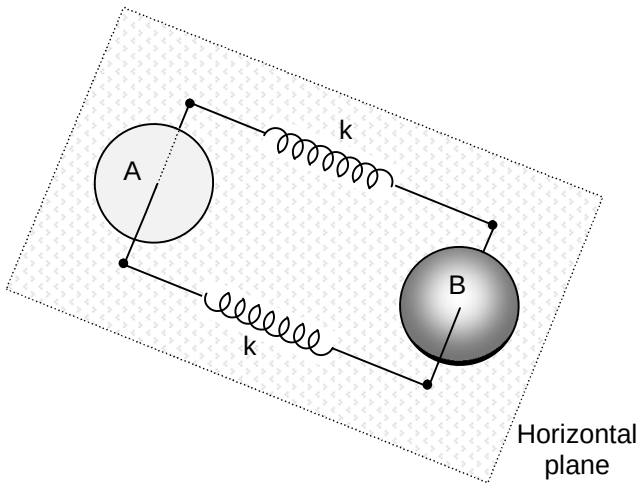



(A) $\frac{12}{5}\frac{mv_0^2}{R}$ (B) $\frac{18}{5}\frac{mv_0^2}{R}$
(C) $\frac{8}{5}\frac{mv_0^2}{R}$ (D) $\frac{5}{4}\frac{mv_0^2}{R}$

## SECTION – A
### (Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

8. A hollow sphere A has mass m and radius R. With the help of two identical massless springs (each of force constant k) it is connected to a solid sphere B having mass m and radius R. The springs are connected symmetrically to the massless axles of the spheres which are passing through the centres of the spheres and being horizontal and parallel to each other as shown in the figure. The spheres are pulled apart so as to stretch each spring by $x_0$ and then released. During subsequent motion, the spheres do not slip.



Match each quantity in List-I with an appropriate value from List-II, and choose the correct option.

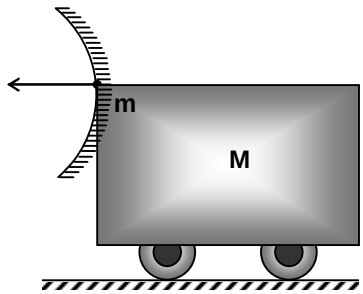
| List –I | | List –II | |
|---|---|---|---|
| (P) | When the spheres are pulled apart so that extension in each spring is $x_0$ and then released. Just after releasing the spheres the magnitude of frictional force acting on the solid sphere is $\frac{P}{7}kx_0$. The value of P is ……. | (1) | 11 |
| (Q) | When the spheres are pulled apart so that extension in each spring is $x_0$ and then released. Just after releasing the spheres the magnitude of acceleration of the centre of mass of the hollow sphere is $\frac{Qkx_0}{5m}$. The value of Q is …….. | (2) | 8 |
| (R) | When the spheres are pulled apart so that extension in each spring is $x_0$ and then released. Just after releasing the spheres The magnitude of acceleration of the centre of mass of the system is $\frac{R\,kx_0}{70m}$ .The value of R is ……. | (3) | 5 |
| (S) | When the spheres are pulled apart so that extension in each spring is $x_0$ and then released. Just after releasing the spheres The magnitude of frictional force acting on the hollow sphere is $\frac{4}{S}kx_0$ . The value of S is …….. | (4) | 6 |
| | | (5) | 4 |

The correct option is:

(A) (P) → (5) (Q) → (4) (R) → (2) (S) → (3) (B) (P) → (3) (Q) → (4) (R) → (2) (S) → (1)

(C) (P) → (5) (Q) → (3) (R) → (1) (S) → (2) (D) (P) → (3) (Q) → (4) (R) → (5) (S) → (1)

9. A tank of mass 450 kg fires a bomb of mass 50 kg with a horizontal velocity 20 m/s relative to the tank. The tank is fitted with a concave mirror of focal length 20 cm facing towards receding bomb. Match the statements from List I with those in List II and select the correct answer using the code given below the lists, when bomb is at a distance 10 cm from the mirror. (assume that gravity free space and neglect the mass of the mirror)




Match each quantity in List-I with an appropriate value from List-II, and choose the correct option.

| List –I | | List –II | |
|---|---|---|---|
| (P) | Magnitude of velocity of Image w.r.t. Pole is ……..m/s. | (1) | 2 |
| (Q) | Magnitude of velocity of Image w.r.t. ground is ……..m/s. | (2) | 100 |
| (R) | Magnitude of velocity of Image w.r.t. object is ……..m/s. | (3) | 82 |
| (S) | Magnitude of velocity of mirror w.r.t. ground is ……..m/s. | (4) | 80 |
| | | (5) | 252 |

The correct option is:

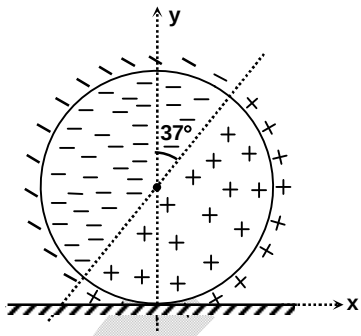
(A) (P) → (4) (Q) → (3) (R) → (2) (S) → (1)

(B) (P) → (4) (Q) → (2) (R) → (5) (S) → (3)

(C) (P) → (3) (Q) → (5) (R) → (2) (S) → (4)

(D) (P) → (3) (Q) → (5) (R) → (4) (S) → (1)

10. A non conducting uniform solid sphere of mass 4kg and radius 2m is having uniform charge density +7 C/m$^3$ on one half and $-7$ C/m$^3$ on another half as shown in figure. It is placed on a rough non conducting horizontal plane. At t = 0 a uniform electric field $\vec{E} = -20\hat{j}$ N/C is switched on and the solid sphere starts rolling without slipping.



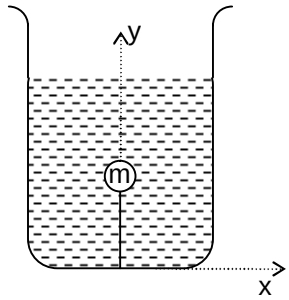

Match each quantity in List-I with an appropriate value from List-II, and choose the correct option.

| List –I | | List –II | |
|---|---|---|---|
| (P) | The acceleration of the topmost point of the solid sphere at t = 0, is X$\pi$ m/s$^2$. The value of X is …… | (1) | 320 |
| (Q) | The magnitude of frictional force at t = 0 is Y$\pi$ Newton. The value of Y is …… | (2) | 224 |
| (R) | Dipole moment associated by the solid sphere is Z$\pi$ Coulomb-meter. The value of Z is …… | (3) | 160 |
| (S) | Torque due to the electric field about the contact point is 4T$\pi$ Newton-meter. The value of T is …… | (4) | 112 |
| | | (5) | 56 |

The correct option is:

(A) (P) $\rightarrow$ (3) (Q) $\rightarrow$ (5) (R) $\rightarrow$ (2) (S) $\rightarrow$ (3)
(B) (P) $\rightarrow$ (4) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (5) (S) $\rightarrow$ (3)
(C) (P) $\rightarrow$ (3) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (5) (S) $\rightarrow$ (2)
(D) (P) $\rightarrow$ (3) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (4) (S) $\rightarrow$ (1)

11. A very large container is filled with a liquid of density $\rho = 1\times10^3$ kg/m$^3$ and a spherical ball of mass m is in equilibrium with the help of a string of length 9 meter as shown in the figure. The density of ball is $\rho/2$ and its volume is $2\times10^{-3}$ m$^3$. Container starts accelerating with acceleration $\vec{a} = \left(4.8\hat{i} - 3.6\hat{j}\right)$ m/s$^2$. (Take g = 10 m/s$^2$ and tan 37° = 3/4)




Match each quantity in List-I with an appropriate value from List-II, and choose the correct option.

| List –I | | List –II | |
|---|---|---|---|
| (P) | Magnitude of tension in the string in equilibrium with respect to the container is $\frac{K}{4}$ newton. The value of K is | (1) | 37 |
| (Q) | The angle made by the string in equilibrium with respect to the container with the vertical is P degree. The value of P is | (2) | 33 |
| (R) | When the ball is in equilibrium with respect to the container, ball is slightly displaced ($\theta \rightarrow 0$) the released, the time period of | (3) | 53 |

| | | | |
|---|---|---|---|
| | oscillation is $\frac{\lambda\pi}{16\sqrt{2}}$ second. The value of $\lambda$ is | | |
| (S) | The angle made by the upper surface of the liquid with the vertical is Q degree. The value of Q is | (4) | 32 |
| | | (5) | 48 |

The correct option is:

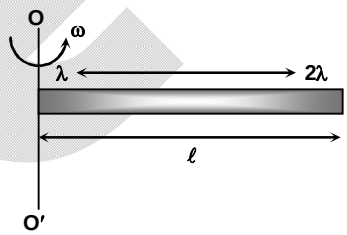
(A) (P) → (3) (Q) → (5) (R) → (2) (S) → (3)
(B) (P) → (4) (Q) → (1) (R) → (5) (S) → (3)
(C) (P) → (4) (Q) → (1) (R) → (5) (S) → (2)
(D) (P) → (3) (Q) → (1) (R) → (4) (S) → (1)

## SECTION – B

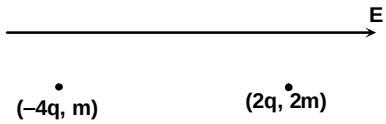
### (Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

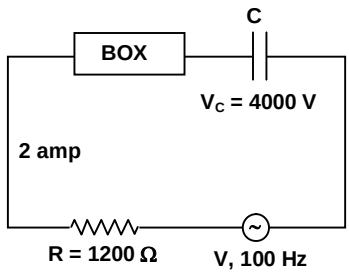
12. A rod of length '$\ell$' can rotate freely in a horizontal plane about vertical axis OO′ passing through one of its ends with angular velocity $\omega$. The mass per unit length linearly increases from $\lambda$ to 2 $\lambda$ from the axis of rotation to other end as shown in the figure. The total extension in the rod is $\Delta\ell \times 10^{-5}$ meter. Find the value of $\Delta\ell$. (take $\omega$ = 2 rad/sec, $\lambda$ = 4 kg/m, $\ell$ = 4 meter, Young modulus Y = $4 \times 10^{11}$ N/m$^2$, cross-sectional area of the rod A = $\frac{7}{3}$ mm$^2$ and neglect the gravity)



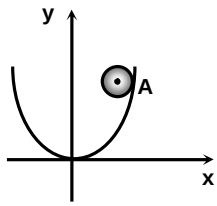

13. Two charged particles (2m, 2q) and (m, –4q) are placed in a gravity free space where a uniform electric field E exists as shown in figure. After the particles are released they stay at a constant distance from each other. The interaction energy between the two charge particle is $-x$ Joule. Find the value of x. (Neglected the gravitational interaction). (k = $\frac{1}{4\pi\varepsilon_0}$ is electrostatic constant and take $kq^3E = \frac{12}{5}$)




14. A circuit shown in the figure contains a box having either a capacitor or an inductor and draws 2 ampere current. The power factor of the circuit is 0.6, while current lags behind the voltage. The inductance/capacitance of the box is $\frac{P}{1000}$ H/F. Find the value of P. (take $\pi$ = 3.2)

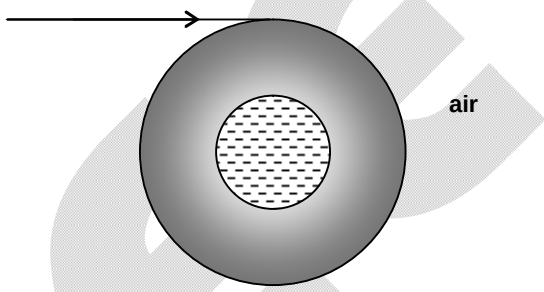


15. A small hollow sphere of mass 8 kg and radius 2 m starts from rest from point A (10m, 5m) on the parabolic surface having sufficient friction such that pure rolling exist after releasing from rest. When the ball arrives at lowest point then normal force is k newton. Find the value of k. (Take g = 10 m/s$^2$)




16. A tube of certain diameter of length 167 cm is open at both ends. Its fundamental frequency of resonance is found to be 100 Hz. The velocity of sound in air is 340 m/s. When one end is open and other end is closed, the frequency of 5$^{th}$ harmonic is P Hz. Find the value of P.

17. A capillary tube made of a material of refractive index $\frac{5}{3}$. The outer radius of the tube is 5m and inner radius is 3m. The tube is filled with a liquid of refractive index $\frac{25}{12}$. A light ray is incident on the tube at grazing incidence (angle of incidence approximately $\approx 90°$) on the outer surface of capillary tube as shown in the figure. The net angle of deviation, when the light ray emerge out from the capillary tube, is k degree. Find the value of k.



# Chemistry

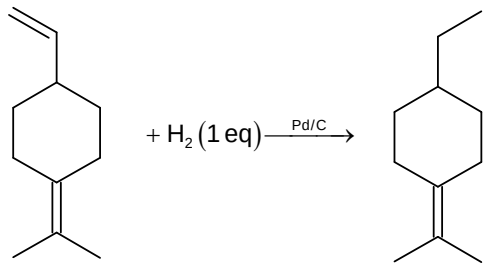
## PART – II

### SECTION – A
**(One or More than one correct type)**

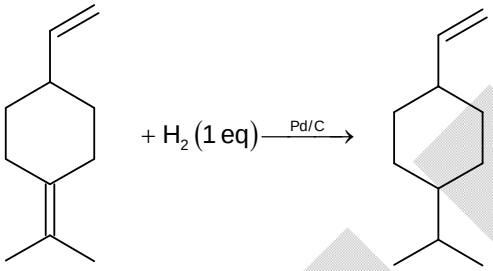
*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

18. The **correct** option is/are

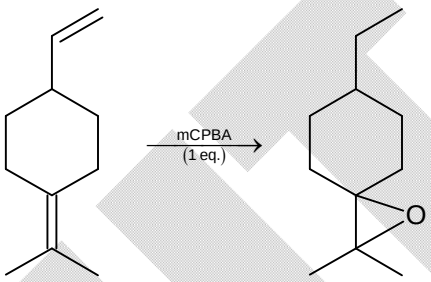
(A)



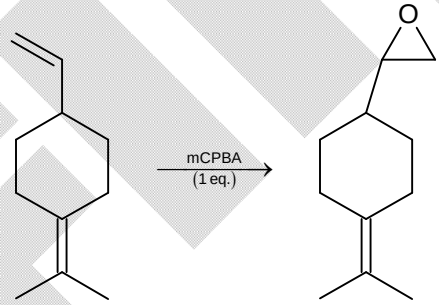

(B)



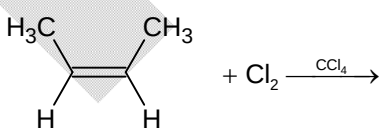

(C)




(D)




19.




Choose correct option is/are for the reaction

(A) Reaction is example of antiaddition

(B) Product mixture is racemic mixture

(C) Number of products formed is 2

(D) Number of product formed is 1

20. Choose **correct** option is/are

(A) for adiabatic irreversible process $\left[\Delta S\right]_{sys} > 0$

(B) for adiabatic irreversible process $\left[\Delta S\right]_{total} > 0$

(C) for adiabatic irreversible process $\left[\Delta S\right]_{surr} = 0$

(D) for spontaneous process at constant temperature and pressure, $\Delta G < 0$

## SECTION – A
### (One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

21. In which complex co-ordination isomerism is not possible?

(A) $\left[Ag(NH_3)_2\right][AgCl_2]$ (B) $\left[Co(NH_3)_6\right]\left[Cr(CN)_6\right]$

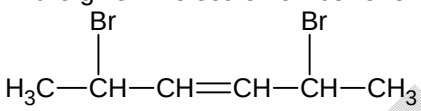
(C) $\left[Cr(NH_3)_6\right]\left[Co(CN)_6\right]$ (D) $\left[Co(NH_3)_6\right]\left[Co(NO_2)_6\right]$

22. In which of the following intensive property is/are

(A) $E_{cell}$ (B) Mass

(C) Internal energy (D) Enthalpy

23. In the given molecule number of enantiomeric pair will be



(A) 2 (B) 4

(C) 3 (D) 1

24. On hydrolysis gives triprotic oxoacid as one of their hydrolysis product

(A) $PCl_5$ (B) $PCl_3$

(C) $BCl_3$ (D) $SO_3$

## SECTION – A
### (Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

25. Match the following

| List – I (Species) | | List – II (Bond order) | |
|---|---|---|---|
| (P) | $O_2^-$ | (1) | 2.5 |
| (Q) | $N_2^+$ | (2) | 1.0 |
| (R) | $H_2^+$ | (3) | S and P mixing not operative |
| (S) | $B_2$ | (4) | 0.5 |
| | | (5) | diamagenetic |

(A) P → 3; Q → 2; R → 5; S → 1

(B) P → 3; Q → 1; R → 4; S → 2

(C) P → 4; Q → 1; R → 5; S → 2

(D) P → 4; Q → 1; R → 2; S → 5

26. Match the following

| List – I (Unit of silicate) | | List – II (Number of shared oxygen atom per tetrahedral unit) | |
|---|---|---|---|
| (P) | $Si_2O_7^{-6}$ | (1) | 3 |
| (Q) | $\left(Si_2O_{5.5}^{-3}\right)_n$ | (2) | 1 |
| (R) | $SiO_2$ | (3) | 2.5 (avg.) |
| (S) | $\left(Si_2O_5^{2-}\right)$ | (4) | 4 |
| | | (5) | 4.5 |

(A) P → 3; Q → 2; R → 4; S → 1
(B) P → 5; Q → 3; R → 4; S → 1
(C) P → 4; Q → 1; R → 5; S → 2
(D) P → 4; Q → 1; R → 2; S → 5

27. Match the following

| List – I (Complex) | | List – II (Number of stereo isomers) | |
|---|---|---|---|
| (P) | $Ma_3b_3$ | (1) | 3 |
| (Q) | $Ma_2b_2c_2$ | (2) | 6 |
| (R) | $Ma_3bcd$ | (3) | 5 |
| (S) | $Ma_3b_2c$ | (4) | 2 |
| | | (5) | 8 |

(Where a, b, c and d are monodentate ligand)
(A) P → 4; Q → 2; R → 3; S → 1
(B) P → 2; Q → 4; R → 3; S → 1
(C) P → 4; Q → 2; R → 1; S → 3
(D) P → 2; Q → 1; R → 4; S → 3

28. Match the following:

| List – I (Molecules) | | List – II (Magnetic property and state of hybridization) | |
|---|---|---|---|
| (P) | $\left[Fe(CN)_6\right]^{4-}$ | (1) | $d^2sp^3$, diamagnetic |
| (Q) | $\left[Fe(CN)_6\right]^{3-}$ | (2) | $d^2sp^3$, paramagnetic |
| (R) | $\left[Cu(NH_3)_4\right]^{2+}$ | (3) | $dsp^2$, paramagnetic |
| (S) | $\left[Ni(Cl)_4\right]^{2-}$ | (4) | $sp^3$, paramagnetic |
| | | (5) | $sp^3d$, paramagnetic |

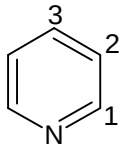
(A) P → 1; Q → 2; R → 3; S → 4
(B) P → 2; Q → 1; R → 3; S → 4
(C) P → 2; Q → 1; R → 4; S → 3
(D) P → 1; Q → 2; R → 4; S → 3

## SECTION – B

**(Numerical Answer Type)**

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

29. Best position for electrophilic approach in




30. Number of dihalogen derivative including stereo isomers for  is

31. An ideal gas at temperature $T_1$ is heated at constant pressure to $T_2$, and then reversibly expanded, according to $PV^x$ = constant, until the temperature is once again $T_1$, if the required value of x is 'P' for the change of entropy during the separate process are equal, $\left[\text{Take}\frac{C_{P,M}}{C_{V,m}}=\frac{3}{2}\right]$ the value 10P is

32. Number of black colour sulphide among the following is
CdS, SnS, NiS, CoS, MnS, PbS, $SnS_2$, $As_2S_3$, CuS

33. Minimum number of hydrogen atom to get maximum number of different spectral lines for the transition n = 4 to n = 1 is

34. The mass of ice that will be separated out on cooling a solution containing 40 g of ethylene glycol in 400 gram of water to $-5^{\circ}C$ is. $\left(K_f \text{ for water} = 1.86 \text{ K mol}^{-1} \text{ kg}\right)$

# Mathematics

## PART – III

### SECTION – A
### (One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

35. Let g: $R^+ \to R$ be a function such that $g(x) = x^2$ and let $h_1(x) = g(x)$ and $h_{(k+1)}(x) = \min_{0 \le t \le x} \left(h_k(t) + g(x-t)\right)$ where $k \in N$, then

(A) $4h_4(x) = \frac{x^2}{4}$ (B) $h_2'(x) = x$

(C) $\sum_{n=1}^{10} \frac{1}{h_n(x)} = \frac{55}{x^2}$ (D) $h_3'(x) = x^2$

36. If $0 \le x, y \le 2\pi$ such that $\tan^4 x + \tan^4 y + 2\cot^2 x \cot^2 y = 3 + \sin^2(x + y)$, then $\cos^2(x - y)$ CAN'T be equal to

(A) 1 (B) 0

(C) $\frac{1}{2}$ (D) $\frac{1}{4}$

37. If the equation $(5\sin x - 12\cos x)^2 - (k^2 + k + 5)|5\sin x - 12\cos x| + k^3 + 3k^2 + 2k + 6 = 0$ has a real solution, then k can be

(A) 12 (B) 9

(C) 0 (D) –4

### SECTION – A
### (One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

38. If c is a point at which Rolle's theorem holds for the function $f(x) = \log\left(\frac{x^2 + k}{8x}\right)$ in the interval [1, 5] where k is a real number, then $f'(c) + f''(c)$ is equal to

(A) $\frac{7}{10}$ (B) $\frac{1}{5}$

(C) $-\frac{1}{5}$ (D) $\frac{3}{20}$

39. Let f: $Z \to Z$ be a function that satisfies $f(x) + f(y) = f(x + y) - xy$ and if $f(5) = 10$, then (Z is the set of integers)

(A) f(x) is one-one and onto (B) f(x) is one-one and into

(C) f(x) is many one and onto (D) f(x) is many one and into

40. In the polynomial function $f(x) = (x - 1)(x^2 - 2)(x^3 - 2^2) \ldots (x^{10} - 2^9)$ the coefficient of $x^{49}$ is

(A) 56 (B) –8

(C) 8 (D) 24

41. Number of integers between 1 and 100000 which have sum of digits equal to 15 is

(A) 3246 (B) 630

(C) 3876 (D) 1850

## SECTION – A
### (Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

42. Match the following List-I with List-II

| List – I | | List - II | |
|---|---|---|---|
| (P) | If $f(x)=\int_0^1 \frac{1}{1+\lvert x-t\rvert}dt$, $x \in (0, 1)$, then $f'\left(\frac{1}{2}\right)$ is equal to | (1) | 2 |
| (Q) | f : [0, 1] $\to$ R be a continuous function such that $\int_0^1 f^2(x)dx=\int_0^1 f^3(x)dx=\int_0^1 f^4(x)dx$, then number of such functions are | (2) | infinite |
| (R) | $\int_0^2 f^4(x)dx=\int_0^2 6xf^2(x)dx-24$, then $f^2\left(\frac{1}{3}\right)$ is equal to | (3) | 0 |
| (S) | $\left[\int_0^\infty \left[\frac{4}{x^2+1}\right]dx\right]$ is equal to (where [.] represents greatest integer function) | (4) | 1 |
| | | (5) | 3 |

The correct option is:
(A) (P) $\to$ (3); (Q) $\to$ (4); (R) $\to$ (1); (S) $\to$ (5)
(B) (P) $\to$ (3); (Q) $\to$ (1); (R) $\to$ (4); (S) $\to$ (5)
(C) (P) $\to$ (4); (Q) $\to$ (1); (R) $\to$ (3); (S) $\to$ (5)
(D) (P) $\to$ (4); (Q) $\to$ (2); (R) $\to$ (3); (S) $\to$ (1)

43. Match the following List-I with List-II

| List – I | | List – II | |
|---|---|---|---|
| (P) | If $(\sin^2\theta_1+\text{cosec}^2\theta_1)^2+(\cos^2\theta_1+\sec^2\theta_1)^2=12+\frac{\sin\theta_2}{2}$. If $\theta_1 \in (0, 2\pi)-\left\{\frac{\pi}{2}, \pi, \frac{3\pi}{2}\right\}$ and $\theta_2 \in (0, 4\pi)$, then number of ordered pairs $(\theta_1, \theta_2)$ is | (1) | 1 |
| (Q) | $2\sin^2\left(\frac{x}{3}\right)\cos^2\left(\frac{x}{9}\right)=x^2+\frac{1}{x^2}$, then number of solution of this equation is | (2) | 24 |
| (R) | If $k=\sum_{n=1}^{8}\int_0^{\frac{\pi}{2}}(1-\sin t)^n \sin 2t\, dt$, then $\frac{5k}{4}$ is equal to | (3) | 8 |
| (S) | If $\alpha$ and $\beta$ are two different values of $\theta$ lying between 0 and $2\pi$ which satisfy the equation $6\cos\theta+8\sin\theta=9$, then the value of $25\sin(\alpha+\beta)$ is | (4) | 0 |
| | | (5) | 2 |

The correct option is:
(A) (P) $\to$ (5); (Q) $\to$ (4); (R) $\to$ (3); (S) $\to$ (1)
(B) (P) $\to$ (2); (Q) $\to$ (4); (R) $\to$ (1); (S) $\to$ (3)
(C) (P) $\to$ (3); (Q) $\to$ (4); (R) $\to$ (1); (S) $\to$ (2)
(D) (P) $\to$ (3); (Q) $\to$ (5); (R) $\to$ (1); (S) $\to$ (2)

44. Match the following List-I with List-II

| List – I | | List – II | |
|---|---|---|---|
| (P) | If the tangents drawn from any point on the circle $x^2 + y^2 - 2hx - 4y + 3 + h^2 = 0$ to the hyperbola $15x^2 - 16y^2 - 90x + 64y - 169 = 0$ are perpendicular, then h is equal to | (1) | 4 |
| (Q) | Number of points on the rectangular hyperbola $xy = c^2$ from which two perpendicular tangents can be drawn to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (where $0 < b < a < c$) is | (2) | Infinite |
| (R) | If the tangents drawn from point P(h, k) to the circle $x^2 + y^2 - 2x - 7 = 0$ are perpendicular, then P moves on curve $C_1$. If the tangents drawn from point P(h, k) to the ellipse $\frac{(x-7)^2}{a^2} + \frac{(y-8)^2}{20} = 1$ are perpendicular the locus of point P is $C_2$. If $C_1$ and $C_2$ have exactly 3 common tangents, then a is equal to | (3) | 0 |
| (S) | If the tangents drawn from the point $(\lambda, 2)$ to the parabola $y^2 - 16x - 6y + 89$ are perpendicular, then $\lambda$ is equal to | (4) | 1 |
| | | (5) | 3 |

The correct option is:
(A) (P) $\rightarrow$ (1); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (4); (S) $\rightarrow$ (3)
(B) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (1); (S) $\rightarrow$ (4)
(C) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (2); (S) $\rightarrow$ (4)
(D) (P) $\rightarrow$ (1); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (1); (S) $\rightarrow$ (3)

45. Consider the given data with frequency distribution

| $x_i$ | 2 | 7 | 9 | 5 | 3 |
|---|---|---|---|---|---|
| $f_i$ | 4 | 8 | 3 | k | 3 |

Match each entry is List-I with entries in List-II

| List – I | | List – II | |
|---|---|---|---|
| (P) | If k = 7, then mean of the above data is | (1) | 1.75 |
| (Q) | If the mean of the above data is 5.5, then median is | (2) | 5.4 |
| (R) | If k = 2, then mean deviation about mean of the above data is | (3) | 2.25 |
| (S) | If the mean of the above data is 5.25, then mode of above data is | (4) | 5 |
| | | (5) | 7 |

The correct option is:
(A) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (5); (R) $\rightarrow$ (3); (S) $\rightarrow$ (4)
(B) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (5); (S) $\rightarrow$ (4)
(C) (P) $\rightarrow$ (4); (Q) $\rightarrow$ (5); (R) $\rightarrow$ (3); (S) $\rightarrow$ (5)
(D) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (5)

## SECTION – B

### (Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

46. The number of ways to choose 6 distinct natural numbers from first 120 natural numbers such that any two chosen numbers differ by atleast 8 can be expressed as ${}^{n}C_{6}$, then n is equal to

47. Let $I_1 = \int_0^1 x^{\frac{5}{2}}(1-x)^{\frac{7}{2}}dx$ and $I_2 = \int_0^1 \frac{x^{\frac{5}{2}}(1-x)^{\frac{7}{2}}}{(3+x)^8}dx$. If $\frac{I_1}{I_2} = 2^m 3^k$, then m + 2k is equal to

48. If the line 3x – 4y + k = 0 ($k \in Z$) lies between the circles $x^2 + y^2 – 2x – 4y – 4 = 0$ and $x^2 + y^2 + 14x – 6y + 57 = 0$ without intersecting or touching either circles, then how many values of k are possible?

49. Let f be a continuous function satisfying the equation $\int_0^x f(t)dt + \int_0^x tf(x-t)dt = e^{-x} - 1$, then $e^{2024}f(2024)$ is

50. Let $p_1, p_2, p_3, ....., p_{12}$ be 12 observations such that $\sum_{i=1}^{12} p_1 = 72$ and $\sum_{\forall\, k<j} p_k p_j = 2328$. Then the variance of $p_1, p_2, p_3, ....., p_{12}$ is equal to

51. If ${}^{60}C_6 + 6\times {}^{60}C_7 + 15\times {}^{60}C_8 + 20\times {}^{60}C_9 + 15\times {}^{60}C_{10} + 6\times {}^{60}C_{11} + {}^{60}C_{12} = {}^{k}C_p$ (Where $p < \frac{k}{2}$), then k + p is equal to